Objectives:

The objectives of this assignment are:

1. To design a fire protection system for KHAN JAHAN ALI HALL, KUET (South building, 1st & 2nd floor),
2. To describe the apparatus required to design a conventional fire protection system for a residential building,
3. To estimate the cost of the fire protection system,
4. To justify the design according to the BNBC & NFPA guidelines, and
5. To learn how to design a professional fire protection system in AUTOCAD environment.

Theory:

A fire protection system is a combination of engineering measures, building design strategies, and mechanical equipment intended to prevent fire occurrence, detect fire at an early stage, control fire spread, and ensure safe evacuation of occupants. Modern fire protection design follows internationally recognized standards such as those developed by the National Fire Protection Association (NFPA) and national regulations such as the Bangladesh National Building Code (BNBC). These standards define the requirements for building construction, fire detection systems, suppression equipment, emergency exits, and firefighting facilities. In the BNBC, fire safety requirements are presented in Part 4: Fire Protection, which covers general provisions, precautionary requirements, means of escape, fire protection equipment, and special requirements for different building occupancies

Objectives of Fire Protection Systems:

The primary objectives of a building's fire protection system include:

i. Preventing fire ignition through safe design and materials
ii. Detecting fire at the earliest stage
iii. Alerting occupants and emergency responders
iv. Controlling and suppressing the fire
v. Providing safe evacuation routes
vi. Reducing structural damage and loss of life

Fire protection engineering is generally divided into two types

1. passive fire protection 2. active fire protection

Passive fire protection refers to the structural and architectural features of a building that limit the spread of fire and smoke. These include fire-resistant walls, floors, doors, and compartmentation within the building. Fire-resistant construction materials and structural

separation help confine fire within a limited area and maintain structural stability for a sufficient time during a fire emergency.

Active fire protection systems involve mechanical or electrical equipment that detects, alarms, or suppresses fires.

Fire Detection Systems:

Early fire detection is crucial because fire growth is exponential, and quick detection significantly reduces damage. Fire detection systems include:

i. Smoke detectors :- Detect combustion particles in the air
ii. Heat detectors :- Respond to temperature rise
iii. Flame detectors :- Detect radiation emitted by flames
iv. Gas detectors :- Detect different gas molecules

Fire Alarm Systems:

Fire alarm systems notify occupants when a fire is detected. Main components include:

I. Fire alarm control panel
II. Manual call points
III. Alarm bells or sirens
IV. Strobe lights
V. Communication systems

Fire Suppression Systems:

Fire suppression systems are used to control or extinguish fires. Components include

I. Automatic Sprinkler System: Sprinklers activate when temperature reaches a preset level. Water is discharged directly onto the fire
II. Fire Hydrant System: Water storage tank, Fire pump, Hydrant valves, Hose reels
III. Portable Fire Extinguishers: Class A (Solid materials), Class B (Flammable liquids), Class C (Electrical fires), Class D (Metal fires), Class K (Cooking oils)

Means of Escape:

Means of escape refers to the safe path that occupants follow to exit a building during an emergency. A properly designed means of escape includes corridors, staircases, exits, and emergency doors that lead to a safe open space outside the building. The BNBC code specifies requirements such as maximum travel distance, minimum exit width, number of exits, emergency lighting, and exit signage to ensure that occupants can evacuate safely without congestion.

Staircases and escape routes must remain unobstructed and be constructed with fire-resistant materials to maintain accessibility during a fire event.

In addition to escape routes, the code requires buildings to be equipped with firefighting facilities and built-in equipment. These include fire hydrant systems, dedicated fire water storage tanks, fire pumps, and hose connections located at strategic points throughout the building. High-rise buildings may also require fire lifts, smoke control systems, and pressurized staircases to facilitate evacuation and firefighting operations. Adequate water supply for firefighting is another critical requirement. Water storage tanks and pumping systems must be designed to deliver sufficient pressure and flow rate to hydrants, sprinklers, and hose reels during fire emergencies.

Furthermore, fire protection design must consider building occupancy and hazard level. Different types of buildings such as residential buildings, hospitals, industrial facilities, and commercial complexes have different fire risks. Therefore, BNBC specifies additional safety measures depending on the occupancy classification, including enhanced detection systems, additional exits, and specialized suppression equipment for high-risk environments.

Components & Ratings:

Photoelectric Smoke Detector:

Model: 3392

Operating Voltage: DC9V-35V

Standby Current: ≤10uA

Alarm Current: ≤10mA

Alarm Sound Level: ≥80 dB

Remote Control Output: ≤15mA

Alarm Indicator: Red LED

Operating Temperature: -10°C to +50°C

Ambient Humidity: ≤95% RH

Coverage Area: 100 m² (space height 10m-10m)

Size: 12x6 cm (4.72x2.36 inches)

Fig.1: Photoelectric Smoke Alarm Fire Detector

Manual Call Point:

- 2 level control mode by a lock
- 24VDC, and 5VDC power supply output
- fire and fault signal output,
- one circuit detector input.

Fig.2: Conventional Manual Call Point

Fire Alarm Strobe Siren:

- Up to100dB alarm pressure
- 20mA working current
- 12v DC power supply
- Three tones

Fig.3: Fire Alarm Strobe Siren

Fire Alarm Control Panel (FACP):

- 8-zones
- Style-B
- 1.25 amps of NAC power
- 24 Volt DC power supply
- 3 modes: silence, non-silence and auto-silence
- program duration: 5 to 20 minutes

Fig.4: 8-zone Fire Alarm Control Panel

Fire Extinguisher:

- Fire Class: ABCE
- Capacity: 5 kg
- Working Pressure: Maximum 14 bar
- Material: 50 % of DCP materials grade
- Working temperature: -30 to +60 degrees
- Testing Pressure: 25 bar
- Refill: Yes
- Mono ammonium phosphate material

Fig.5: ABCE Dry Powder 5 Kg Fire Extinguisher

Fire Sprinkler head:

- Made of: Steel
- Type: Sprinkler, wet, pendent
- Product: Tyco upright sprinkler quick response
- Water Pressure: 175 PSI
- Working temperature: 68^0c

Fig.6: Sprinkler head

Cable:

Detector Loop:

- Cross-section: 1.5 mm² (2-core)
- Voltage rating: 300/500 V (or 450/750 V)
- Conductor: Annealed stranded copper (Class 2)
- Insulation & Sheath: FRLS PVC (Fire Retardant Low Smoke, zero halogen)
- Flame retardant standard: IEC 60332-1
- Current carrying capacity (in conduit): 15–18 A
- Low-current 24 V DC supervised Class A loop for smoke detectors and Manual Call Points

Sounder Circuit cable:

- Cross-section: 2.5 mm² (2-core)
- Voltage rating: 300/500 V (or 450/750 V)
- Conductor: Annealed stranded copper (Class 2)
- Insulation & Sheath: FRLS PVC (Fire Retardant Low Smoke, zero halogen)
- Flame retardant standard: IEC 60332-1
- Current carrying capacity (in conduit): 20–24 A
- Higher-current circuit for hooters/horns

Fig.7: BRB FRLS 2 core cable

Sprinkler pipe, valves & accessories:

Piping (GI Schedule-40):

- Branch lines (to each sprinkler head): 1 inch (25 mm)
- Cross mains (along corridors & veranda): 1.25 inch (32 mm)
- Vertical Riser (at stairs connecting both floors + pump): 2 inch (50 mm)

Valves & Accessories:

- Zone Control Valve (with supervisory switch)
- Alarm Check Valve (with water flow switch)
- Inspector's Test Valve & Connection
- Main Drain Valve
- Pressure Gauges
- Tamper Switches

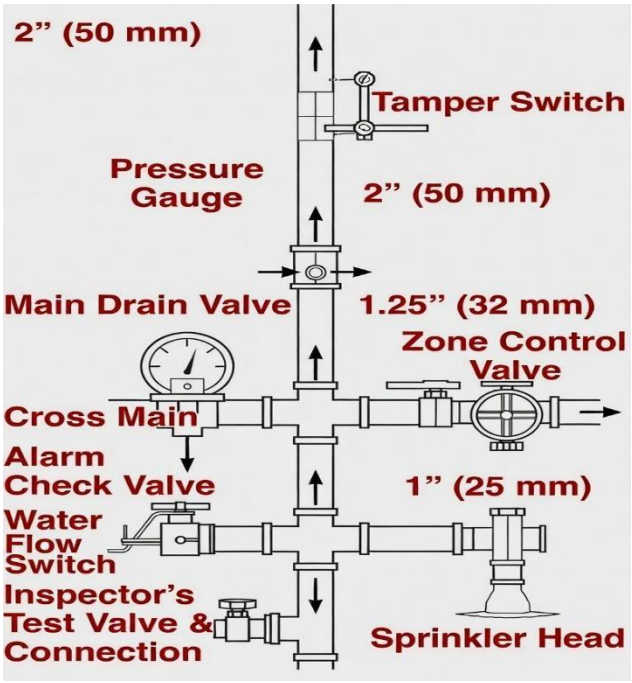


Fig.8: Sprinkler pipe, valves & accessories

Cost Estimation:

Table 1: Cost Estimation Table for the Fire Alarm system

Item	Quantity (For Two floors)	Unit price (BDT approx.)	Total (BDT)
Smoke Detector (Light Scattering)	44 pcs	414	18,216
Manual Call Point (Break Glass)	6 pcs	600	3,600
Fire Alarm (Strobe Siren)	42 pcs	1,040	43,680
FACP (8 zone) + 24v Battery Bank	1 pc	38,000	38,000
Fire Extinguisher (5 KG, Type: ABC, Dry Powder)	14 pcs	1,350	18,900
Sprinkler Head	156 pcs	250	39,000
Detector Loop Cable (1.5 mm^2 FRLS, 2/4 core, BRB Cable)	1200 m (Approx.)	3,250 per 100m	39,000
Sounder Circuit (2.5 mm^2 FRLS, 2 core, BRB Cable)	700 m (Approx.)	6,150 per 100m	43,050
Sprinkler Piping (with Valves & accessories)	600m (Approx.)	_	2,00,000 (Approx.)
			Total= 4,43,446

Standard Engineering Approximation:

Installation & Labor = 15% Testing

& Commissioning = 5%

Contingency = 5%

Total Extra= 15+5+5 = 25%

So, Grand total project cost = Material cost + Extra = 4,43,446 + 0.25*4,43,446

= 5,54,307.5 ≈ 5.5 Lakhs BDT

N.B: Conduit & Fitting cost for the cables was not included as it was instructed that it is not necessary to show in the design. The price per unit was taken from online product website bdstall.com

Discussion

The conventional/zonal fire protection system implemented in Khan Jahan Ali Hall South building 1st & 2nd floors (Occupancy A3 – Mess, Boarding House and Hostels) at KUET represents a complete, integrated life-safety solution. The design followed the requirements of BNBC and NFPA standards. The justification of the design is given in this discussion section.

Photoelectric (Light scattering) smoke detectors were used as it suits the residential areas to provide the most accurate detection & least false alarm. It was selected as per BNBC Section 4.4 and Appendix C for their superior performance against smoldering fires typical in residential sleeping areas. Heat detectors respond only after temperature rises & mostly used in kitchen areas; Flame detectors require a direct line-of-sight to flames & used in high risk industrial areas; Gas detectors are required only where flammable gases are used. So, among all the detectors smoke detector was the most suited. With 22 detectors per floor (44 total), each providing a 7.5 m radius coverage (100 m² maximum area), ensuring max 30 ft between two detectors & max 15 ft distance from wall corner. Smoke detectors were mounted on ceilings. In some cases it may seem in the design that the room area is already covered by the corridor smoke detectors but each room needs its separate smoke detector because if the room door is closed smoke may not come out of room. Putting smoke detectors in every room gave that additional safety. No detectors were placed inside bathrooms or showers to avoid false alarms from steam, in full compliance with BNBC guidelines that do not mandate detectors in wet areas.

Manual Call Points (MCP) were located at every exit root & one in the middle of corridor, ensuring no occupant travels more than 100 ft to reach one, satisfying BNBC Section 4.3 and NFPA 72 criteria. The mounting height of the operable part of MCP should be 1.2 m.

Although the smoke detectors used, has in built alarm & LED indicator but strobe siren were still used for additional safety if somehow detector alarm & LED fails to indicate. Strobe siren delivers audible notification exceeding 65 dB throughout the floor with flash indicator in case anyone has hearing problems. The mounting height should be 2.3 to 2.5 m (top of device)

All devices were wired in supervised Class A loops using 1.5 mm²(RM) FRLS copper for detection/MCP circuits and 2.5 mm² (RM) FRLS copper for sounder circuit, precisely the fire-resistant, low-smoke cabling mandated by BNBC Sections 2.7 and 4.3 together with Part 8 electrical provisions.

The central Fire Alarm Control Panel (installed in the ground floor) of 8 zones (Z-2 for 1st & Z-3 for 2nd floor) with 24 V DC battery bank provides primary mains power plus automatic standby supply for 24–72 hours, guaranteeing uninterrupted operation during power failure as required by BNBC Section 4.3 and NFPA 72. The design was a conventional fire protection system where each floor was sectorized as a zone. FACP can understand in which zone fire/smoke has been detected. But

it can't precisely tell which detector was activated. That can be done in addressable FACP where each device has its own unique address. In conventional systems if a detector senses something the detector loop current changes that is sensed by FACP & it turns on all the alarms connected in series. That's why FACP is called the brain of fire protection system as all the sensed data go to FACP & then it takes control of the system. In addition, Interface Unit (IFU) could be connected to the FACP which could control the entire building's existing system (Ex: elevators, generators, motors etc.) in case of an emergency.

Automatic suppression was achieved through a complete wet-pipe sprinkler system compliant with BNBC Section 4.2.4. Sprinkler heads were installed at a maximum 15 ft spacing (20 m^2 per head), using 1" (25 mm) branch lines, 1.25" (32 mm) cross mains, and 2" (50 mm) vertical risers. The pipe was connected to zone control valves, alarm check valve, inspector's test connection, and drain valves as required.

Portable fire extinguishers (5 kg ABC dry chemical powder) were placed at every staircase and at intervals not exceeding 30 ft travel distance, fulfilling BNBC Section 4.10 and NFPA 10 requirements for light-hazard buildings. The mounting height should be 1.2 m.

EXIT signs were installed above every staircase door, at veranda ends, and at corridor junctions with directional arrows, following BNBC Section 3.16 and NFPA 101 for clear, continuous means of escape. The mounting height should be 2.1-2.4 m (bottom of sign). Illuminated signs need to be installed.

The overall cost for the two floors was 5.5 Lakhs which seems reasonable in comparison to the market.

The AUTOCAD design of Khan Jahan Ali hall provided by the authority included only the ground floor design ensuring identical design for all floors. So, the fire protection system design is similar for both 1st & 2nd floor. That's why one was shown. The other one will be identical.

Conclusion: Overall, every component in the design has been selected and located in accordance with BNBC and the relevant NFPA standards. The integrated system delivers early detection, automatic suppression, reliable alarm notification, and safe evacuation paths, providing a high level of life safety while remaining economical compliant. This design not only meets but exceeds the minimum code requirements, demonstrating a balanced, professional approach to fire protection system.

References:

[1] Bangladesh National Building Code (BNBC), Appendix C: Detailed Guidelines for Selection and Siting of Fire Detection System, Part 4: Fire Protection. Housing and Building Research Institute (HBRI), Dhaka, Bangladesh, 2006.

[2] National Fire Protection Association, NFPA 13: Standard for the Installation of Sprinkler Systems, NFPA 72: National Fire Alarm and Signaling Code, NFPA 10: Standard for Portable Fire Extinguishers, and NFPA 101: Life Safety Code, Quincy, MA, USA, 2022.

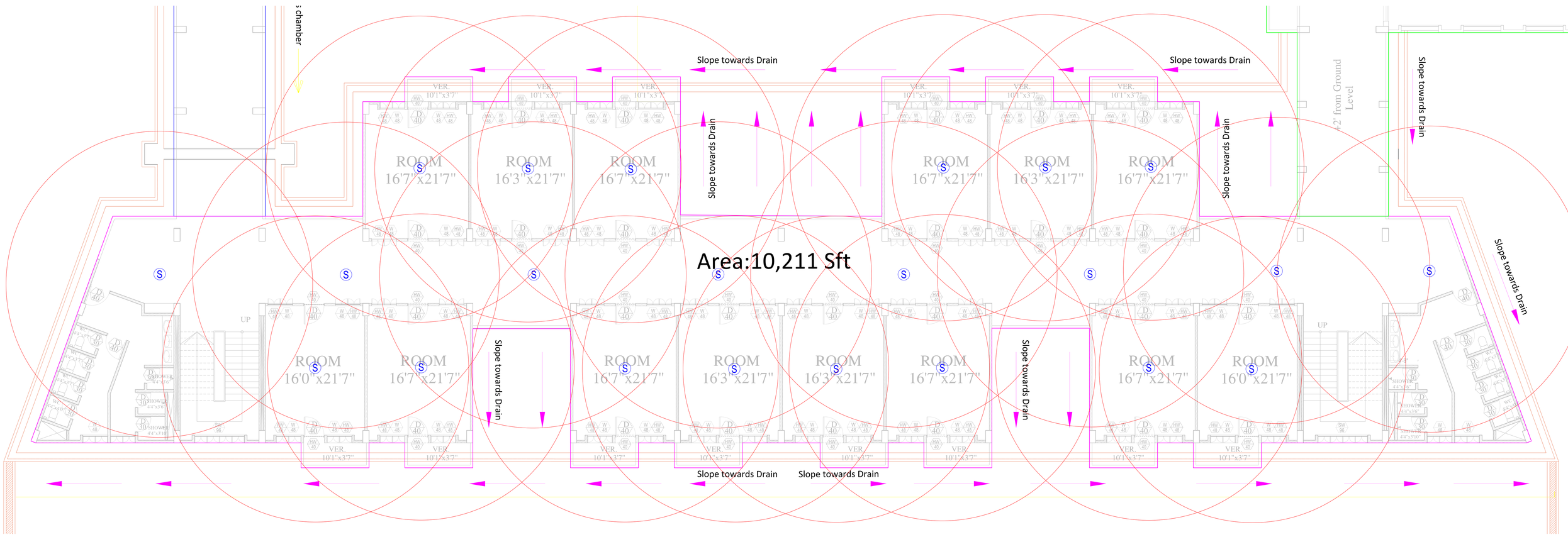
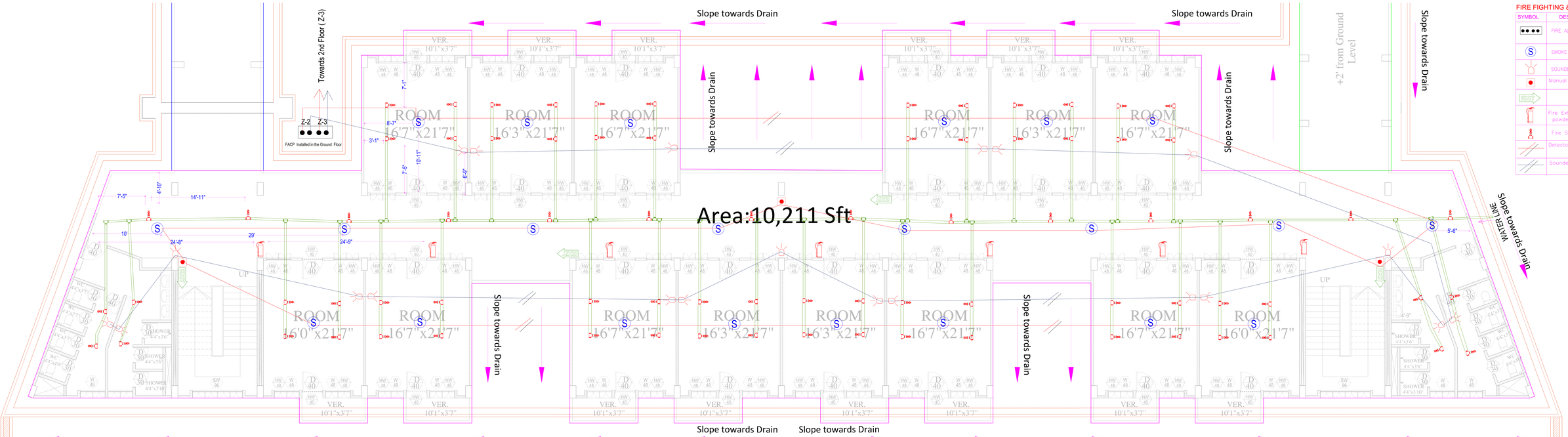


Fig : Area covered by the Smoke Detectors



FIRE FIGHTING & FIRE ALARM SYSTEM

SYMBOL	DESCRIPTION
	FIRE ALARM CONTROL PANEL ZONES/LOOP
S	SMOKE DETECTOR (Light Scattering)
	SOUNDER & STROBE
	Manual Call Point (Break Glass Unit)
EXIT	EXIT sign
	Fire Extinguisher (5kg, Type: ABC Dry chemical powder , 43A : 233B : C)
	Fire Sprinkler Head
	Detector Loop : 1.5 mm² FRLS Copper
	Sounder Circuit: 2.5 mm² FRLS Copper

Fire Protection system

KHAN JAHAN ALI HALL, KUET [South building ,1st & 2nd Floor (Identical)]